

REPORT - ASSIGNMENT 5- APL

Keyboard Optimization Assignment

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In this assignment, we have to optimize the keyboard layout, by moving the characters around to give the minimal possible distance.

In my approach, I first modify the dictionary containing the initial Qwerty layout to contain the position of the home row keys, instead of the characters to prevent any confusion as the characters get moved around. For e.g : instead of 'start' : 'a', I put 'start': (1.75, 2) (initial pos of a). Also added the enter key in the characters dictionary for handling multi-line strings and kept the remaining combinations as they are.

Defined a new distance calculator to account for the changes in the layout dictionaries, and Enter key. Coming to the optimization part, I used simulated annealing by first swapping keys from the initial layout (keys dictionary) and then calculating distance in the new layout and storing the new layout and new distance as a current layout and current distance only if new distance < current distance or if it is based on probability p defined as $p = \text{np.exp}((\text{current_distance} - \text{new_distance}) / \text{temp})$, then if it is better than the current best it gets into the best layout and best distance dictionaries.

I also plot the keyboard layout and distance plot animation which shows the distance decreasing and coming to the optimal minimal distance.

I tried the code on the examples - "hello world" and `'''Hello World!`

```
1234, ~!@#$$%^&*()_+
This is a test of the keyboard layout optimizer.
Line #1: Quick brown fox jumps over the lazy dog.
Line #2: Testing: numbers, symbols, and more...
[Example] -- <HTML> -- (Version 2.0)
"Quotes", 'single quotes', and `backticks`
Newlines, tabs and spaces are here!
End of test.
'''
```

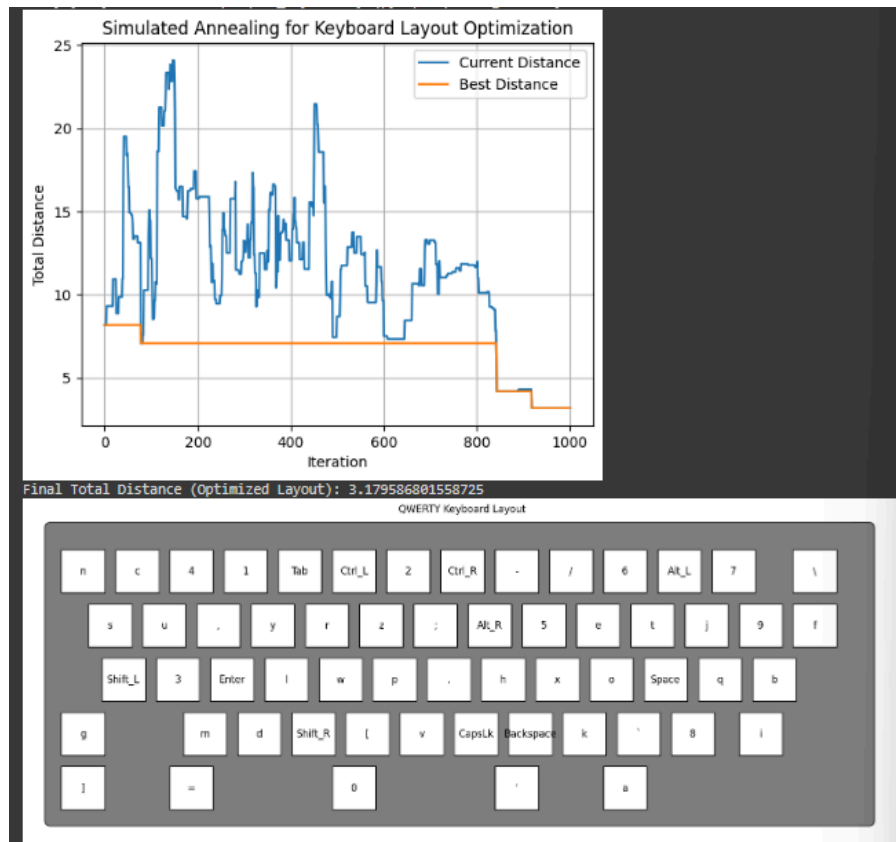
And for hello world, the distance went down from 8.17 to 3.18, for the second example it went from 444 to 232 units, which is a significant decrease.

The two animations are one for distance(using the function - `animate_simulated_annealing`

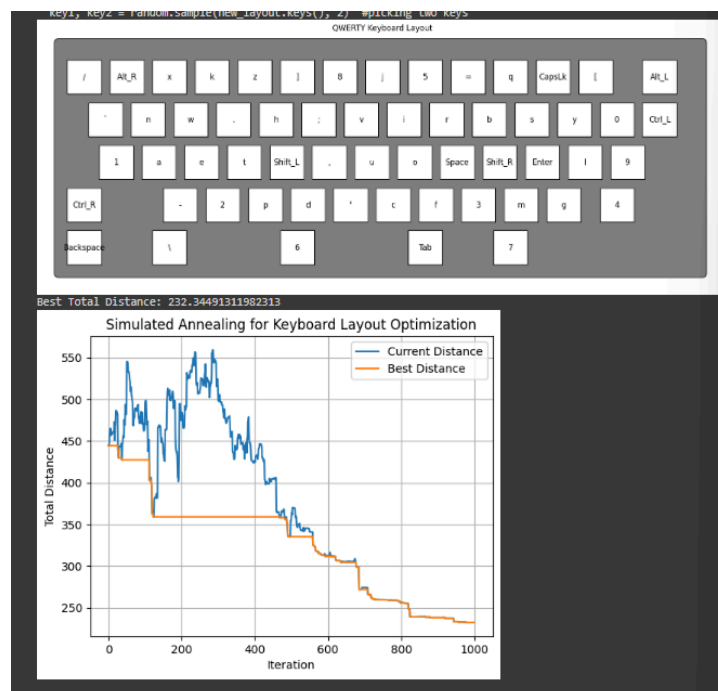
and other for the keyboard animations⁹ along with distance plot animation using the function - `animate_allsimulated_annealing` that shows the keyboard plots of all the best layouts. However the second one takes a longer time to run.

PLOTS:

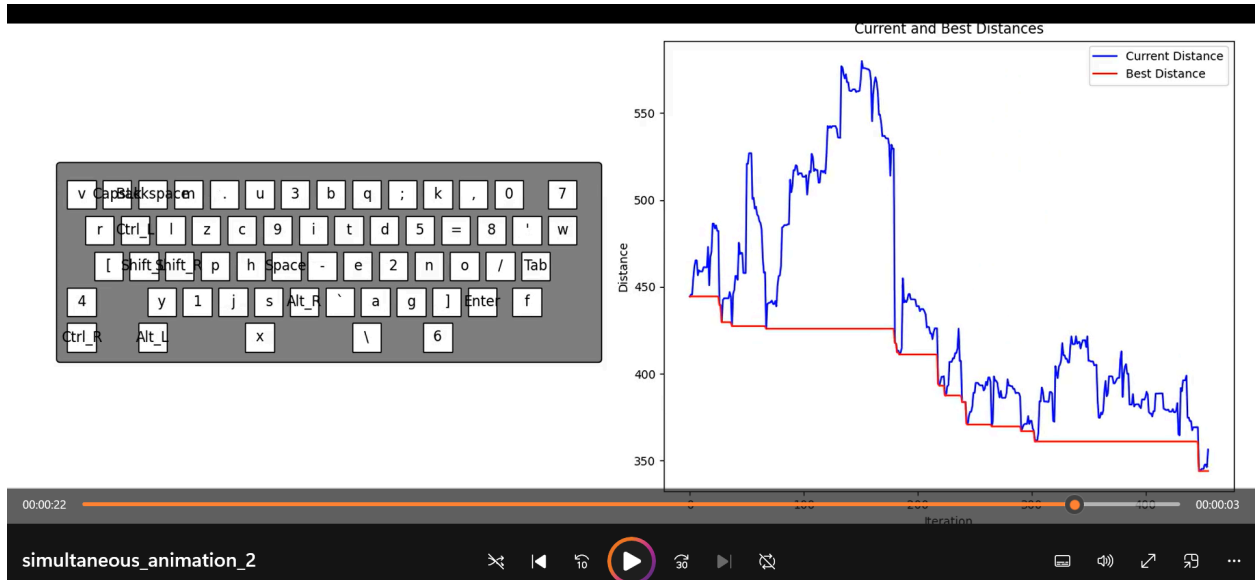
Example 1:



Example 2:



Animation results of example :2: This is hen cooling rate is 0.99



For the first animation I put the cooling rate as 0.995, whereas in my code I put 0.99 as I was trying out different cooling rates. So the animations attached are for 0.995.s

Colab link of the assignment : [🔗 EE23B188- Nishitha.D - Assignment 5 - EE2703](#)

